UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 29128

LH ASTEROSEISMOLOGY & EXOPLANETS

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. Cool main-sequence stars have outer convective envelopes. Hence they show evidence of the surface phenomenon called granulation.

   (a) Provide a brief overview of the mixing-length approach to modelling convection, and how this relates to the granulation. **[8]**

   (b) Let the fractional RMS fluctuation in stellar intensity due to an individual granule be (dI/I), where I is the mean intensity. Derive the following expression for the stellar RMS amplitude of the stellar disc-averaged signal

   $$\sigma = \left(\frac{dI}{I}\right)\left(\frac{x}{R_\star}\right),$$

   where x is the average size of a granule, and $R_\star$ is the stellar radius. Explain any assumptions made in your derivation. **[6]**

   (c) Use the mixing-length approximation to show that the above equation implies that $\sigma \propto T_{\rm eff} R_\star / M_\star$, where $T_{\rm eff}$ is the effective temperature and $M_\star$ the mass of the star. Again, state any assumptions made in the derivation. **[10]**

   (d) On the lower-main sequence, $M_\star \propto T_{\rm eff}^{1.5}$ and $M_\star \propto R_\star$. Hence comment on how you would expect the total photometric power detected in solar-like oscillations to vary with mass in this part of the Hertzsprung-Russell diagram. **[6]**

2. An exoplanet with a mass and a radius equal to those of Jupiter's orbits a Sun-like star ($5777\,\mathrm{K}, 1\,\mathrm{M}_\odot, 1\,\mathrm{R}_\odot$) at an orbital separation of $1.5\ \mathrm{AU}$. This planet is still surrounded by a circumplanetary disc in which satellites are forming. The disc extends from the planet's surface to the edge of the Hill sphere and is composed of spherical ice grains that have a density $\rho = 0.9\,\mathrm{g.cm^{-3}}$, and a radius $0.5\,\mathrm{cm}$.

   (a) Derive from first principles an expression for the Roche radius of a satellite orbiting a planet. In a first step, use that expression to calculate the minimum orbital radius that a forming satellite could occupy. In a second step assume that as it is forming, the satellite increases its density by a factor two. Estimate how much closer it could survive.
   Give your answers in km. **[6]**

   (b) Adapt the equation for a disc's scale-height so it is applicable to an ice disc around a planet, and calculate the height containing 99% of the disc's mass. State clearly any assumption you make, providing an answer in metres. Comment on your answer.
   Knowing that that the full height of Saturn's rings is $10\,\mathrm{m}$ provide some reasons as to where the difference might come from. **[10]**

   (c) The disc has now dissipated, and the satellite has managed to grow to Titan's mass ($1.34 \times 10^{23}\ \mathrm{kg}$, $\rho = 1.8\,\mathrm{g.cm^{-3}}$). Discuss whether this satellite will be detectable by the transit method, supporting your argument with relevant calculations. **[6]**

   (d) As the planet and the satellite orbit their common barycentre, the satellite will cause the planet to transit sometimes early, sometimes late. Assuming a one minute uncertainty on the timing of each transit, how many years are needed to reach a 5σ detection of the perturbations in transit time due to the satellite? [Start by estimating the planet's semimajor axis about the common centre of mass.] State any assumptions you make. **[8]**

3. Consider a $1\,\mathrm{M}_\odot$ star. Its radius changes from $1\,\mathrm{R}_\odot$ on the main sequence to $4\,\mathrm{R}_\odot$ at the base of the red-giant branch. The star has two gas giant planets. The inner planet is on an orbit with eccentricity $e = 0.94$, and semimajor axis $a = 0.7\,\mathrm{AU}$, with a mass $m_\mathrm{p} = 1\,\mathrm{M}_\mathrm{Jup}$. The second planet occupies a circular orbit at $5\,\mathrm{AU}$.

   (a) Explain how, and why, the oscillation spectrum and granulation signal detected in photometric observations change as a $1\,\mathrm{M}_\odot$ star evolves from the main-sequence to become a red-giant star. Your explanation should discuss the underlying physics involved and include relevant equations, as appropriate. **[13]**

   (b) While the host star is on the main sequence, the inner planet's orbit will circularise under the action of tides, and the planet will become a hot Jupiter. Derive a general expression for the planet's final semimajor axis, in terms of its periastron, and use this expression to show that the planet's final semimajor axis is less than $0.1\,\mathrm{AU}$. **[5]**

   (c) The mean surface rotation period of the star is 22 days. Calculate in which direction the inner planet will tidally migrate, explaining the physical process involved. **[4]**

   (d) The timescale for tidal evolution is

$$\tau = 10^9 \left(\frac{a}{0.05\mathrm{AU}}\right)^5 \left(\frac{R_\star}{1\,\mathrm{R}_\odot}\right)^{-5} \left(\frac{M_\star}{1\,\mathrm{M}_\odot}\right) \left(\frac{m_\mathrm{p}}{1\,\mathrm{M}_\mathrm{Jup}}\right)^{-1} \mathrm{years}.$$

   Estimate the tidal timescale when the host is on the main sequence, and when the host is at the base of the red-giant branch. Hence comment on what will happen to the planet in each case. **[4]**

   (e) Following the red-giant phase, the star will evolve to become a white dwarf, having a mass of $0.5\,\mathrm{M}_\odot$. Comment on what will happen to the outer planet. **[4]**

Formula Sheet

Asymptotic relation for oscillation frequencies (Equation 1.24)
$\nu_{nl} = \Delta\nu\,(n + l/2 + \epsilon) - Dl(l+1)$

Limb darkening law (Equation 2.02)
$g(\cos\theta) = I(\theta)/I(0) = 1 - \sum_{k=1}^{N} c_k\,[1 - \cos\theta]^k$

Mode visibility for photometric observations (Equation 2.07)
$S_l = 2\pi\sqrt{2l+1}\int_0^{\pi/2} P_l(\cos\theta)W(\cos\theta)\cos\theta\,\sin\theta\,d\theta$

Mode visibility for Doppler velocity observations (Equation 2.08)
$S_l = 2\pi\sqrt{2l+1}\int_0^{\pi/2} P_l(\cos\theta)W(\cos\theta)\cos^2\theta\,\sin\theta\,d\theta$

Disc-averaged r.m.s. intensity fluctuation due to granulation (Equation 3.09)
$\sigma_{\mathrm{c},I} \simeq (\delta I/I)_{\mathrm{c}}/\sqrt{n}$

Characteristic timescale for granulation (Equation 3.13)
$\tau_{\mathrm{c}} \simeq l_{\mathrm{c}}/v_{\mathrm{c}} \equiv \alpha H_p/v_{\mathrm{c}}$

Power spectral density mode profile (Equations 4.03 and 4.05)
$P(\nu) \propto \frac{|K(\nu)|^2}{\eta^2}\left(1 + \left(\frac{\nu-\nu_0}{\eta/2\pi}\right)^2\right)^{-1} \equiv H\left(1 + \left(\frac{\nu-\nu_0}{\eta/2\pi}\right)^2\right)^{-1}$

Mode linewidth (Equation 4.04)
$\Gamma = \eta/\pi = (\pi\tau)^{-1}$

Mode power (Equation 4.07)
$A^2/2 = A_{\mathrm{rms}}^2 = \pi/2(H\Gamma)$

Orbital Angular momentum
$L = m_{\mathrm{p}}\sqrt{GM_\star a}$

Protoplanetary disc scale height
$H = \sqrt{\frac{kT(a)a^3}{GM_\star \mu m_{\mathrm{p}^+}}}$

Equilibrium Temperature
$T_{\mathrm{eq}} = T_\star(1-A)^{1/4}\sqrt{\frac{R_\star}{2a}}$

The Hill sphere radius
$r_{\mathrm{H}} = a\sqrt[3]{\frac{m_{\mathrm{p}}}{3M_\star}}$

Escape velocity
$v_{\mathrm{esc}} = \sqrt{\frac{2Gm_{\mathrm{p}}}{r_{\mathrm{p}}}}$

Surface gravity
$g = \frac{Gm_{\mathrm{p}}}{r_{\mathrm{p}}^2}$

Kinetic energy of a gas
$E_{\mathrm{K}} = \frac{3}{2}kT$

Mass accretion
$\frac{dm}{dt} = \pi r^2 v\rho\,(1+\Theta)$ with $\Theta = \frac{v_{\mathrm{esc}}^2}{v^2}$

Isolation mass is
$m_{\mathrm{iso}} = 8\frac{\pi^{3/2}}{\sqrt{3}}a^3 M_\star^{-1/2}\Sigma^{3/2}$

Newton's law
$\boldsymbol{F}_{1,2} = m_{1,2}\ddot{\boldsymbol{r}}_{1,2} = \pm\frac{Gm_1m_2}{r^3}\boldsymbol{r}$

Conic's equation
$r = \frac{h^2/\mu}{1+e\cos(\theta-\varpi)}$

Area of an ellipse
$A = \pi ab = \pi a^2\sqrt{1-e^2}$

Orbital angles relations
$\varpi = \Omega + \omega; \theta = f + \varpi$

Specific angular momentum
$h = \sqrt{\mu a(1-e^2)} = L$

Kepler's second law
$\dot{A} = \frac{1}{2}\sqrt{\mu a(1-e^2)}$

Kepler's third law
$P^2 = \frac{4\pi^2}{\mu}a^3$

Radial velocity equation
$V_{\mathrm{rad}} = \frac{2\pi a}{P}\frac{\sin i}{\sqrt{1-e^2}}[\cos(f+\omega) + e\cos\omega] + \gamma$

Motion caused by a planet
$K_\star = \frac{m_{\mathrm{p}}}{M_\star+m_{\mathrm{p}}}\frac{2\pi}{P}\frac{a\sin i}{\sqrt{1-e^2}}$

Planet's mass function
$f(m_{\mathrm{p}}) = \frac{m_{\mathrm{p}}^3\sin^3 i}{(M_\star+m_{\mathrm{p}})^2} = (1-e^2)^{3/2}\frac{PK_\star^3}{2\pi G}$

Transit depth
$D = \frac{r_{\mathrm{p}}^2}{R_\star^2}$

Scaled radius
$\frac{R_\star}{a} = \frac{W}{P}\frac{\pi}{\sqrt{(1+\sqrt{D})^2-b^2)}}\frac{1+e\sin\omega}{\sqrt{1-e^2}}$

Stellar density
$\rho_\star \sim \left(\frac{a}{R_\star}\right)^3 \frac{4\pi^2}{GP^2}$

Planet gravity
$g_\mathrm{p} = \frac{2\pi}{P}\frac{K_\star\sqrt{1-e^2}}{\sin i}\left(\frac{a}{r_\mathrm{p}}\right)^2$

Light deflection angle
$\epsilon = \frac{4GM_\mathrm{L}}{rc^2}$

Einstein radius
$\theta_\mathrm{E} = \sqrt{\frac{4GM_\mathrm{L}}{\mu D_\mathrm{L}c^2}}$ with $\mu = 1 + \frac{D_\mathrm{L}}{D_\mathrm{LS}}$

Microlensing amplitude
$A = \frac{u^2+2}{u\sqrt{u^2+4}}$

Duration of microlensing
$t_\mathrm{E} = \theta_\mathrm{E}/\nu$

Atmospheric scale height
$H = \frac{kT}{\mu m_\mathrm{p}+g}$

Transmission spectroscopy
$D(\lambda) = D + \frac{2r_\mathrm{p}HA_\lambda}{R_\star^2}$

Planck's law
$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda kT)}-1}$

Signal to Noise ratio
$SNR \sim \frac{A}{\sigma_i/\sqrt{N}}$

Type I migration timescale
$\tau_\mathrm{I} = \frac{1}{2}\frac{H/a}{m_\mathrm{p}/M_\star}\frac{M_\star}{\Sigma a^2}\Omega_\mathrm{p}^{-1}$ with $\tau_e \sim 2.6(H/a)^2\tau_\mathrm{I}$ and $\tau_i \sim 3.7(H/a)^2\tau_\mathrm{I}$

Type II migration timescale
$\tau_\mathrm{II} = \frac{a^2}{\alpha H^2\Omega} \times \max\left(1, \frac{m_\mathrm{p}}{4\pi\Sigma a^2}\right)$

Mutual Hill radii
$\Delta r_\mathrm{H} = \frac{a_1+a_2}{2}\left(\frac{m_{\mathrm{p},1}+m_{\mathrm{p},2}}{3M_\star}\right)^{1/3}$

Tidal acceleration
$a_\mathrm{tides} = \pm\frac{2GM_\star\Delta r}{r^3}$

Roche radius

$a_{\mathrm{Roche}} = r_{\mathrm{p}} \sqrt[3]{3 \frac{M_\star}{m_{\mathrm{p}}}}$

tidal circularisation timescale

$\tau_{\mathrm{circ}} = \frac{4}{63} Q \sqrt{\frac{a^3}{G M_\star}} \left(\frac{m_{\mathrm{p}}}{M_\star}\right) \left(\frac{a}{r_{\mathrm{p}}}\right)^5$

Tidal synchronisation timescale

$\tau_{\mathrm{syn}} = Q \left(\frac{r_{\mathrm{p}}^3}{G m_{\mathrm{p}}}\right) \left(\frac{m_{\mathrm{p}}}{M_\star}\right)^2 \left(\frac{a}{r_{\mathrm{p}}}\right)^6 (\omega_{\mathrm{p}} - \omega_{\mathrm{syn}})$

Tidal infall timescale

$\tau_{\mathrm{infall}} \propto \left(\frac{a}{R_\star}\right)^5 \left(\frac{M_\star}{m_{\mathrm{p}}}\right)$

Rossiter-McLaughlin effect amplitude

$K_{\mathrm{RM}} = \frac{2}{3} D v \sin i_\star \sqrt{1 - b^2}$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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